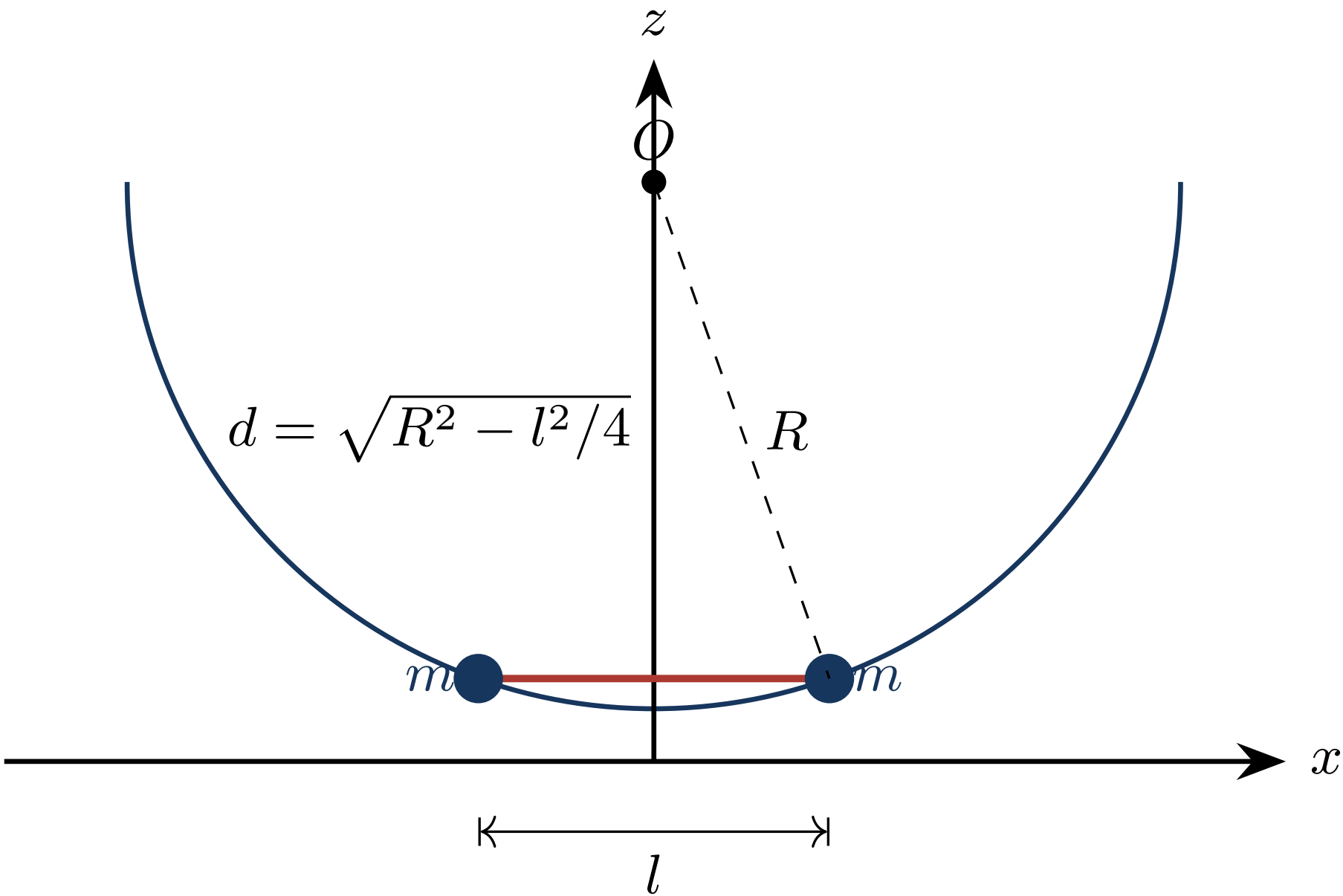
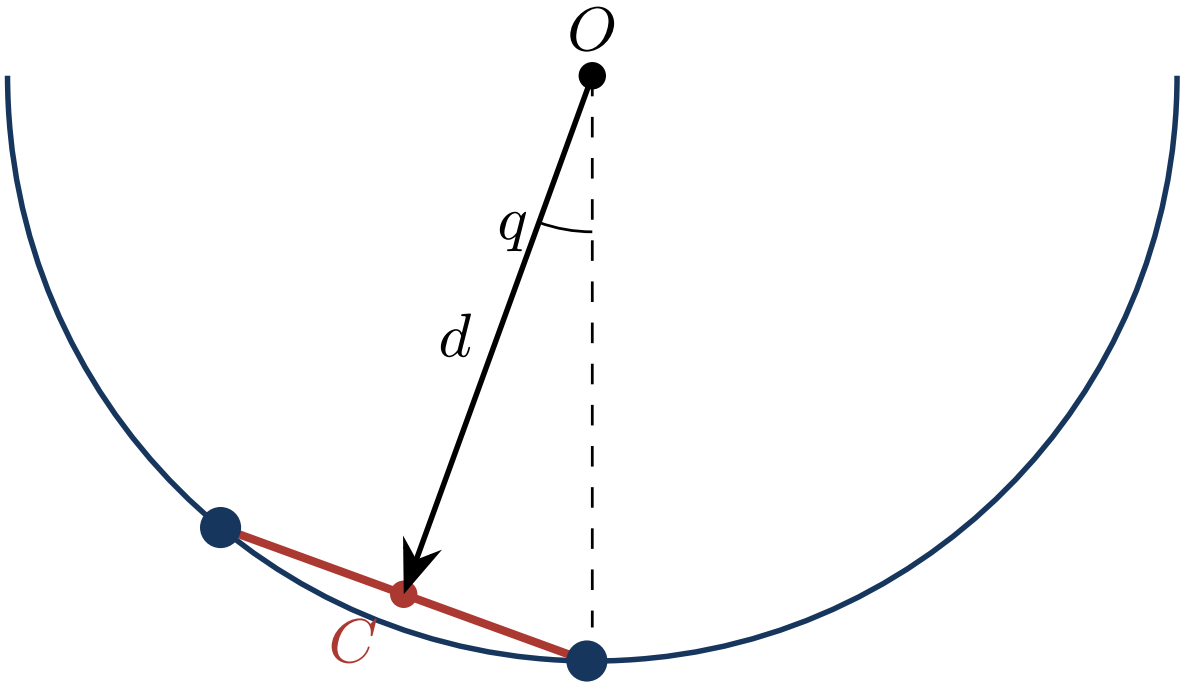
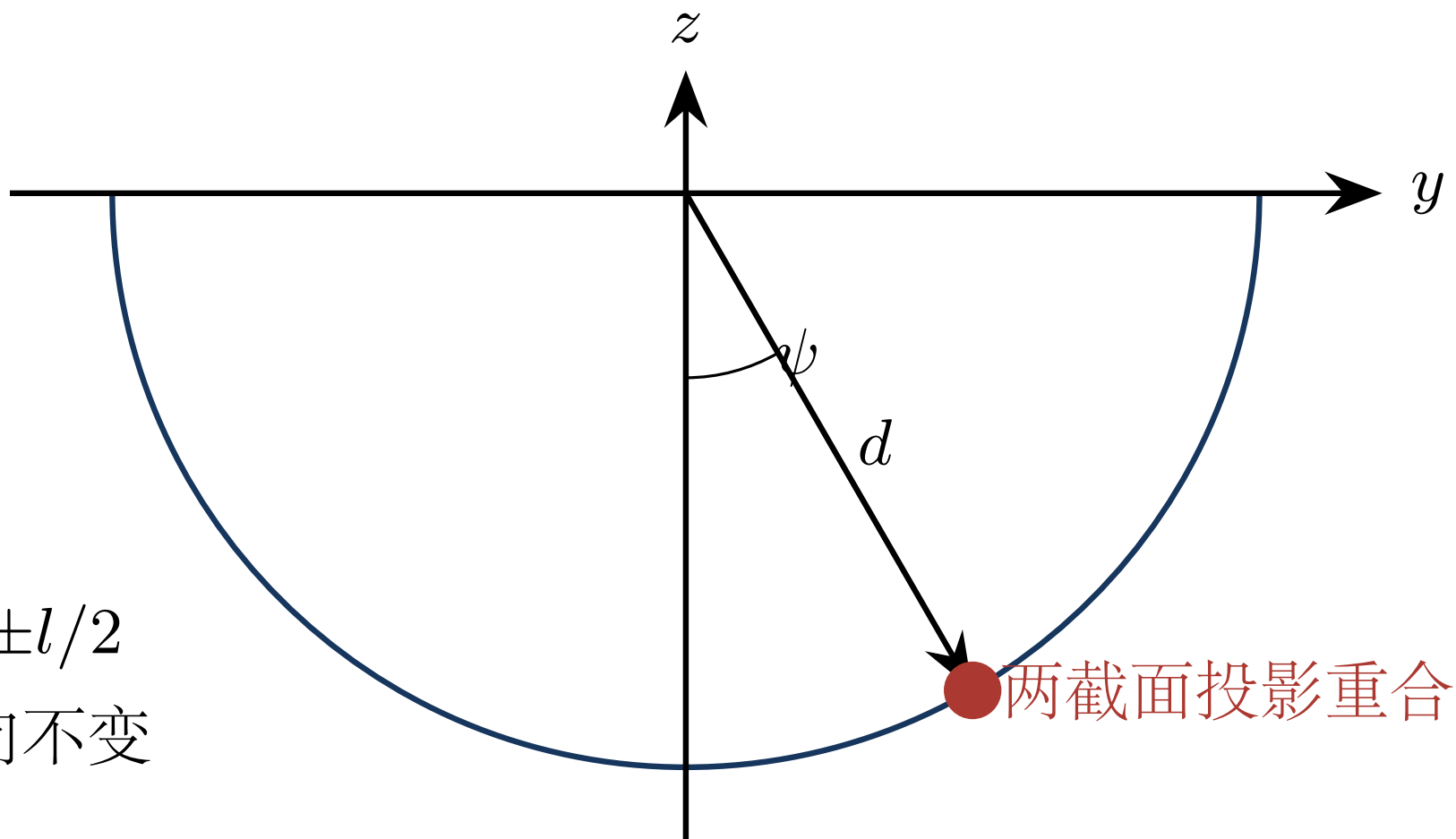

z
O
$d = \sqrt{R^2 - l^2/4}$
R
m
m
x
l



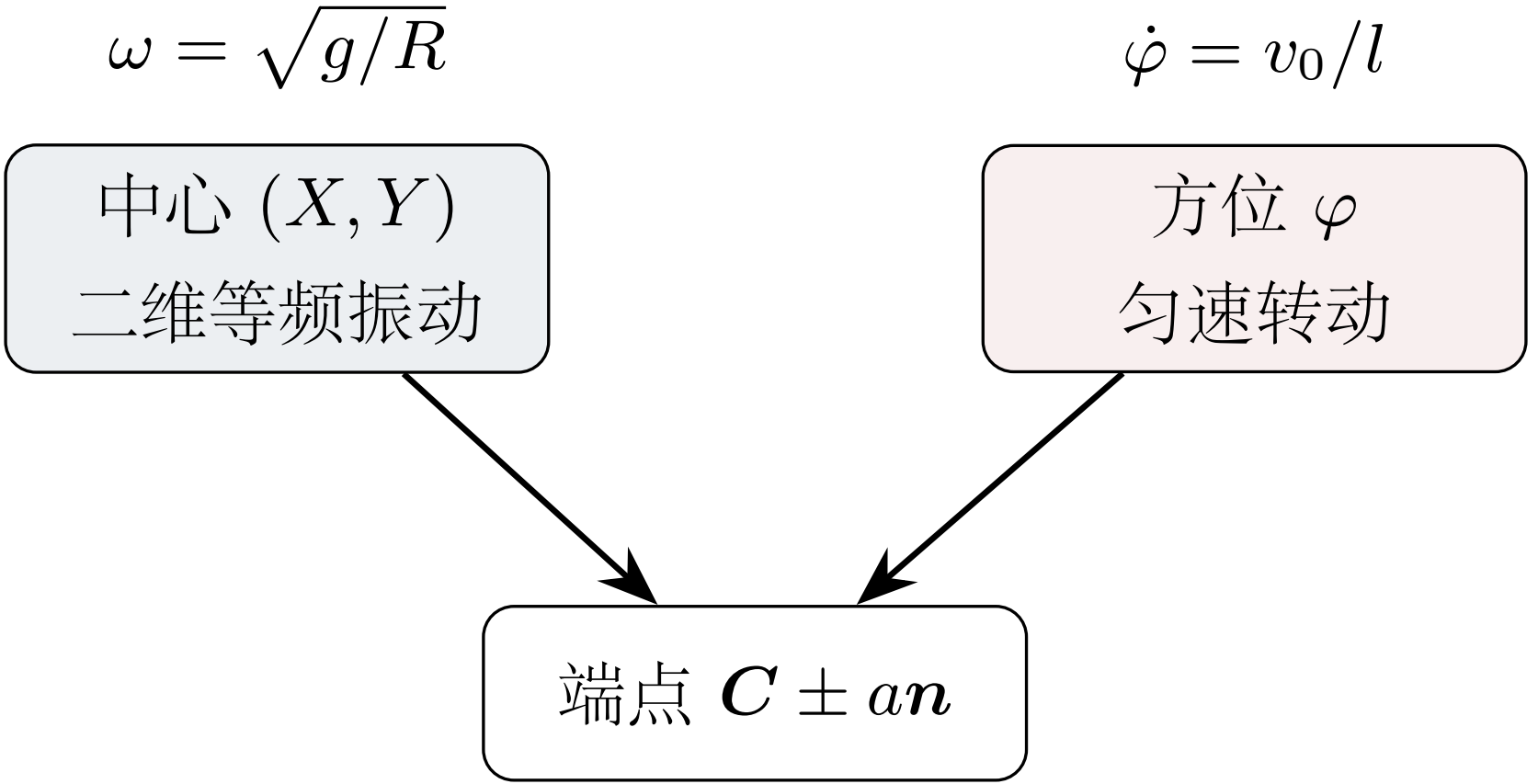
$$K = mR^2\dot{q}^2$$

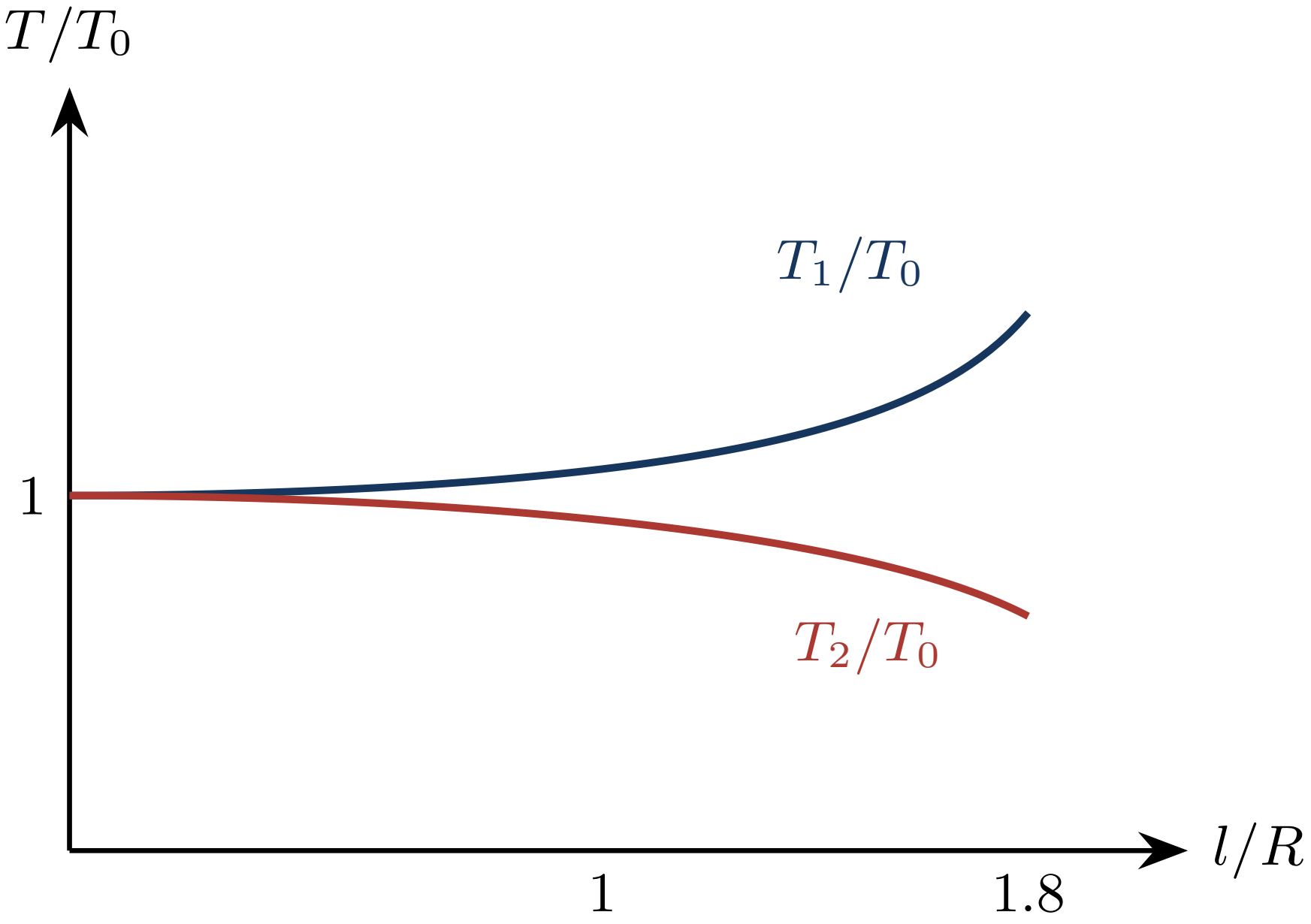
$$U = 2mgd(1 - \cos q)$$



$x = \pm l/2$

杆方向不变


$\omega = \sqrt{g/R}$
$\dot{\varphi} = v_0/l$
中心 (X, Y)
二维等频振动
方位 φ
匀速转动
端点 $\boldsymbol{C} \pm a\boldsymbol{n}$



$$T_1 T_2 = T_0^2$$